

ResidueLink

Smart Stubble-to-Biomass Matching Marketplace | Team ResidueLink |
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1. Summary

Problem Statement

Preventing crop stubble burning by providing farmers an automated, predictive, and profitable marketplace connecting them with biomass buyers and collection machinery.

The Issue

After harvesting paddy, farmers in Punjab and Haryana have only a narrow 10–20 day window to clear their fields before sowing wheat. Burning the leftover stubble is the fastest and cheapest method, making it a primary cause of severe winter smog across Delhi-NCR. While detection systems exist, farmers still lack a timely, reliable, and financially viable platform to schedule collection machinery and connect with verified buyers before the burning window closes.

Our Solution

ResidueLink is a bilingual (Hindi/Punjabi), mobile-friendly web platform that automates the stubble trade lifecycle using predictive harvest modeling and distance-optimized marketplace matching.

- **Predictive & Manual Harvest Scheduling (Core Feature):** At onboarding, farmers input their farm size, crop variety, sowing date, and local parameters. Combined with real-time agro-climatic data from a Weather API, an ML regression model predicts the harvest date. The platform automatically creates a pre-harvest listing and alerts nearby biomass buyers **10 days prior** to the harvest. Farmers also have the flexibility to manually set or override their harvest date.
- **Proximity-First Marketplace:** Buyer notifications and matches are prioritized by **shortest travel distance** to minimize freight expenses and logistical delays.
- **Negotiation & Pricing:** Biomass buyers post buy requests with their offering rate per ton. Once matched, both parties can communicate directly to negotiate and finalize rates.
- **Mutual Profitability Framework:** A built-in cost-benefit estimator factors in farmer expenses (baler rental, machine operation, diesel) against the buyer's offer and logistical freight costs to ensure a profitable deal for both sides.

2. Detailed Key Features

- **Smart Harvest Prediction & Automated Listing (Core ML Feature):** During onboarding, farmers input baseline data such as farm size, crop variety, and sowing date. The backend integrates with weather APIs (Open-Meteo/OpenWeatherMap) to fetch real-time agro-climatic conditions. An ML regression model analyzes this tabular dataset to predict the exact field maturity date. **10 days prior to this predicted date**, the platform automatically generates a pre-harvest "Sell Request," giving buyers ample time to secure logistics before the urgent burning window opens.
- **Flexible Manual Override & Direct Listing:** Recognizing that field conditions can change unpredictably, farmers are not locked into the ML prediction. They possess full flexibility to manually override the predicted harvest date, adjust timelines, or post instant "Sell Requests" if they have already harvested their fields.
- **Least-Distance Priority Matching & Notifications:** To minimize heavy freight expenses, the platform operates on a proximity-first algorithm using Distance Matrix APIs. When a farm's stubble listing goes live, the system instantly sends push notifications to verified biomass buyers within the vicinity, prioritizing matches with the shortest geographical travel distance.
- **Buyer-Led Bidding & Direct Negotiation System:** The marketplace supports two-way interaction. Buyers can post aggregate "Buy Requests" detailing their total tonnage requirements and their proposed rate per ton. Once matched with a farmer's supply, both parties can communicate directly to negotiate terms, accommodating the flexible nature of local agricultural trade.
- **Mutual Profitability Framework & Dynamic Calculator:** To guarantee that deals are financially viable, the platform features a transparent, built-in profitability calculator. Farmers input their operational expenses (CHC baler rental, machine operation, and diesel costs), while buyers input their transport and logistics overhead. The UI instantly calculates the estimated net profit for *both* sides based on the negotiated rate per ton, ensuring fair trade.

3. Dashboard-Wise Features

Farmer Dashboard

- **Farm Profile & Crop Input:** Interface to enter farm size, crop type/variety, sowing date, and farm location.
- **ML-Predicted Harvest Date:** Displays the automatically calculated harvest maturity date based on crop parameters and live weather API data.
- **Automated 10-Day Pre-Harvest Listing:** System-generated pre-listing that activates 10 days prior to the predicted harvest date, alerting nearby buyers in advance.
- **Manual Date Override & Instant Sell Request:** Option to manually set the harvest date, adjust timelines, or post an immediate sell listing with available tonnage.
- **Baler Availability Status:** Real-time visibility into whether collection balers are available at local Custom Hiring Centres (CHCs).
- **Incoming Offers & Buyer Matches:** Displays interested biomass buyers ranked by least geographical distance.
- **Farmer Profitability Calculator:** Built-in tool to calculate net earnings per acre/ton by factoring in the buyer's offered rate per ton, baler rental costs, machine operator charges, and tractor diesel expenses.
- **Direct Negotiation & Communication Panel:** Direct contact channels (click-to-call or chat) to negotiate and finalize pricing with matched buyers.

Biomass Buyer Dashboard

- **Tonnage Requirement Posting:** Create "Buy Requests" specifying required stubble volume (in tons) and the proposed starting price per ton.
- **Proximity-Based Farm Alerts:** Push notifications for newly listed stubble within the procurement radius, prioritized by shortest distance.
- **Distance & Logistics Tracker:** View farm locations and calculate freight distance to streamline tractor/truck routing.
- **Offer & Deal Negotiation Management:** Review incoming farm listings, submit counter-offers, and access farmer contact details upon acceptance.
- **Buyer Profitability & Landed Cost Calculator:** Evaluates total raw material procurement cost per ton by combining the negotiated price per ton, transport freight & vehicle diesel costs, and loading/unloading handling charges.
- **Procurement & Delivery Tracker:** Manage confirmed pickups, active supply contracts, and completed tonnage deliveries.

Government / CHC Official Dashboard (Prototype)

- **Baler Machinery Status Manager:** Interface for Custom Hiring Centres (CHCs) to update the operational status and availability of stubble-baling machines.
- **Regional Stubble Supply & Demand Heatmap:** Spatial view of active sell requests and buyer demand clusters across districts to monitor harvest activity.
- **Machinery Allocation Overview:** High-level inventory tracking showing where CRM-subsidized balers are currently stationed or assigned.
- **Environmental Impact Analytics:** Aggregated metrics tracking total tons of stubble diverted to industrial use and estimated crop fires prevented.

4. Technical Stack

Frontend (Website)	React.js (responsive web app), HTML5, CSS3, JavaScript
UI / Design	Canva / Figma
Backend / API	Python (Flask / FastAPI)
External APIs	Open-Meteo / OpenWeatherMap API (Agro-climatic weather parameters), Distance Matrix API
Machine Learning	Scikit-learn, Pandas, NumPy (Regression/Tabular models for harvest prediction)
Database	MySQL via phpMyAdmin / PostgreSQL
Demo & Prototyping	Streamlit
Deployment & DevOps	Git/GitHub, Docker, Render / Vercel

5. Feasibility & Viability

- **Feasibility:** By eliminating complex satellite image processing and focusing on tabular crop data and standard weather APIs, the platform is reliable, fast, and fully achievable within a hackathon timeframe.
- **Viability:** ResidueLink aligns with government-backed Custom Hiring Centre (CHC) schemes and the expanding biomass power/pellet manufacturing industry, providing commercial and regulatory value from day one.

6. Impact & Benefits

- **Environmental:** Substantially curtails field fires, mitigating winter smog in North India and preserving soil organic carbon and microbial health.
- **Economic:** Converts crop residue into an additional income stream for farmers while lowering raw material procurement overheads for bio-energy plants.
- **Operational:** Eliminates last-minute logistical bottlenecks by giving buyers and machinery operators a 10-day advance planning window.

7. Loopholes & Future Scope

Open Gaps

- Baler machinery supply during peak season remains constrained across certain clusters.
- Initial adoption requires field-level outreach to onboard local biomass buyers and farmers.

Future Scope

- **Live CHC Machine Tracking:** Upgrading the Government Official portal to reflect live, GPS-integrated baler statuses and automated dispatching.
- **Baler Route Optimization:** Cluster-based routing algorithms for baler operators to service adjacent farms in a single trip, cutting equipment transit fuel.
- **In-App Trust & Rating System:** Reputational scoring for verified buyers and farmers based on timely pickups, moisture levels, and payment settlements.